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Nordgren PINNs to VQE: Advancing Hydraulic Fracturing Simulations in Shale Reservoirs

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ABSTRACT

This study advances hydraulic fracturing simulations in shale reservoirs using two computational paradigms, Physics-Informed Neural Networks (PINNs) and the Variational Quantum Eigensolver (VQE). PINNs were employed to solve Nordgren's equation, which governs fracture width evolution, by embedding physical laws into the neural network architecture. Using TensorFlow on Google Colab, the PINN training process incorporated Adam, L-BFGS, and Newton-CG optimizers, reaching a final loss of 1.23×10^{-9} for a representative benchmark case (e.g., fluid viscosity $\mu = 8.0 \text{ Pa} \cdot \text{s}$, fracture height $H = 2.0 \text{ m}$, and leak-off coefficient $C_L = 1.0$); these parameters are used for comparison and are not tied to a specific field dataset. However, this approach demands computational resources, with training times exceeding 1454 s and memory usage of 1136 MB. Conversely, the VQE framework leveraged Qiskit on Qbraid to optimize the Hamiltonian representing the fracture system in practice. With qubit-based ansatz circuits and classical optimizers (SPSA, COBYLA, and L-BFGS), VQE achieves energy minimization, converging to $-0.583 + 0j$ in under 2 s with low memory requirements. Spatiotemporal fracture width predictions from VQE align with trends but exhibit slight oscillations due to quantum noise. This comparative study highlights a trade-off: PINNs show stronger physics-consistent accuracy in the tested cases, while VQE provides lower runtime and memory use. These results suggest that combining PINN accuracy with VQE speed may be useful for decision-support workflows in hydraulic fracturing. Future research will explore integrating these paradigms for scalable, high-fidelity simulations across complex geological settings.

1 | Introduction

Hydraulic fracturing [1, 2], commonly referred to as fracking, is a cornerstone of modern energy production, enabling the extrac-

tion of hydrocarbons from low-permeability shale reservoirs. This process involves injecting high-pressure fluid into the reservoir to create fractures, allowing oil and gas to flow [3–5]. Proppants, granular materials used to keep these fractures open, play a

Abbreviations: Adam, Adaptive Moment Estimation; COBYLA, Constrained Optimization By Linear Approximation; HF, Hydraulic fracturing; HHL, Harrow-Hassidim-Lloyd; H, Hamiltonian; L-BFGS, Limited-memory Broyden-Fletcher-Goldfarb-Shanno; NNCG, Newton conjugate gradient; PINNs, Physics-Informed Neural Networks; PDEs, Partial Differential Equations; RAR, Residual-based Adaptive Refinement; SPSA, Simultaneous Perturbation Stochastic Approximation; VQE, Variational Quantum Eigensolver.

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critical role in maintaining sustained hydrocarbon flow [6]. The design and properties of proppants—such as size, shape, crush resistance, and interaction with reservoir fluids and stresses—are pivotal to the efficiency of hydraulic fracturing. Poorly optimized proppants risk being crushed under high pressures, leading to fracture closure and diminished productivity [7].

To address these challenges, innovations in proppant materials [8] and mesh designs have advanced hydraulic fracturing operations. High-pressure, high-temperature (HPHT) sintered bauxite particles now provide superior resistance to pressures exceeding 19,000 psi, while multi-mesh particle distributions optimize both sealing and fracture conductivity [9]. Despite these developments, accurately predicting fracture behavior under diverse reservoir conditions remains a key hurdle, requiring precise computational models that balance complexity and efficiency.

Traditional numerical approaches, such as finite element and finite difference methods, have long been the backbone of hydraulic fracturing simulations [9, 10]. However, these methods are computationally intensive, particularly for high-dimensional parameter spaces and heterogeneous reservoir conditions. As a result, researchers have turned to Physics-Informed Neural Networks (PINNs) [11–15], which incorporate governing physical laws into machine learning frameworks [16, 17]. PINNs excel at solving forward and inverse problems by minimizing the residuals of partial differential equations (PDEs), such as Nordgren’s model, which governs fracture width evolution. These networks offer promising capabilities for parameter inference, including estimating fluid leak-off coefficients and proppant settling velocities. However, their computational cost, scalability issues, and sensitivity to large-scale domain heterogeneity highlight the need for complementary methods.

Quantum computing [18, 19] emerges as a promising complementary paradigm in this context. Quantum algorithms, such as the Variational Quantum Eigensolver (VQE) [20] and Harrow-Hassidim-Lloyd (HHL) [21] algorithm, may accelerate selected high-dimensional computations under suitable problem structures and hardware conditions. Combined with hybrid classical-quantum frameworks, these methods can simulate large-scale hydraulic fracturing problems with enhanced efficiency and scalability. For instance, forward problems like predicting fracture width and inverse problems such as parameter estimation can leverage quantum computing’s ability to optimize complex multi-parameter systems.

This study addresses the limitations of classical approaches by exploring the transition from PINNs to quantum computing algorithms for hydraulic fracturing simulations. It presents a physics-informed framework-based on Nordgren’s model [22] to train PINNs for forward and inverse problems and evaluates their performance against quantum solvers. The proposed hybrid computational paradigm integrates the strengths of PINNs with the scalability of quantum solvers, providing a comprehensive methodology to improve the accuracy, computational efficiency, and scalability of hydraulic fracturing simulations. The framework is intended to support hydrocarbon recovery analysis in challenging reservoir conditions and may also be extensible to geothermal energy, carbon sequestration, and other subsurface flow problems.

2 | Problem and Computational Setup

The study investigates the computational challenges of modeling fracture dynamics in hydraulic systems, specifically focusing on Nordgren’s equation as a governing framework. Nordgren’s equation describes the evolution of fracture width in a fluid-driven fracture, incorporating physical parameters such as the Young’s modulus of a homogeneous shale matrix, fluid viscosity, fracture height, and leak-off effects. Accurate and efficient computation of fracture dynamics is critical for optimizing real-world engineering applications, such as hydraulic fracturing, where computational complexity arises from coupling physical constraints with temporal and spatial dependencies.

The computational framework for this study involves leveraging two distinct paradigms: physics-informed neural networks (PINNs) for data-driven modeling and variational quantum eigensolvers (VQEs) for quantum-inspired optimization. Each paradigm addresses different facets of the problem and complements the other when assessing convergence behavior and solution quality for fracture simulations.

2.1 | PINN Development With TensorFlow on Google Colab

For PINN-based simulations, the TensorFlow library [23] was used to define and train neural network architectures that encode the physical constraints of Nordgren’s equation into the loss function. PINNs allow direct optimization of the solution by penalizing deviations from the governing equations and boundary conditions. Using the Google Colab platform [24] provided an accessible, cloud-based computational environment with GPU acceleration, significantly reducing training time. The PINN was trained using a combination of Adam [25], L-BFGS [26], and Newton-CG optimizers [27], allowing for both gradient-based optimization and fine-tuning of weights. The flexibility of TensorFlow facilitated the definition of custom loss functions incorporating physical residuals, enabling precise control over model convergence and boundary adherence.

2.1.1 | Training Configuration

Full-batch training was used with $N = 5000$ collocation points uniformly sampled in $(x, t) \in [0, 1] \times [0, 1]$ per epoch, and $N_b = 2$ boundary points at $x = \{0, 1\}$ with $w = 0$. The boundary weight was fixed at $\lambda_{bc} = 30$. Adam was run for 1000 epochs with an exponential learning-rate decay (initial 10^{-3} , decay_steps = 500, decay_rate = 0.9), followed by L-BFGS-B and Newton-CG fine-tuning via SciPy. No explicit L2 regularization or dropout was applied; the physics and boundary losses act as implicit regularizers and convergence was monitored against analytical/implicit solutions.

2.2 | VQE Development With Qiskit on Qbraid

For VQE computations, the Qiskit framework was employed on the Qbraid platform to explore the quantum-inspired optimization of the fracture Hamiltonian. The VQE algorithm was

designed to minimize the energy of the Hamiltonian, mapping the fracture dynamics onto a quantum computational framework. Qbraid [28] provided a streamlined platform for deploying Qiskit [29] with minimal setup overhead, enabling efficient execution of the VQE workflow. Optimizers such as SPSA [30], COBYLA [31], and L-BFGS were implemented to explore convergence characteristics, with specific attention to their computational time and memory usage. The statevector simulator backend in Qiskit provided a classical emulation of quantum states, supporting controlled benchmarking against analytical and PINN solutions.

3 | PINNs for Hydraulic Fracturing

Physics-Informed Neural Networks (PINNs) provide a modern computational framework for solving partial differential equations (PDEs) by embedding physical laws directly into the training process of a neural network. Unlike traditional machine learning methods that rely heavily on data [32, 33], PINNs leverage the governing equations of the system to enforce physical consistency, reducing dependence on large datasets [34]. In hydraulic fracturing (HF), PINNs are applied to simulate fracture dynamics by solving Nordgren's model for fracture width evolution and fluid transport.

3.1 | Governing Equations for Hydraulic Fracturing

The basis of this work is Nordgren's equation, which describes the evolution of fracture width $w_0(x, t)$ over space x and time t . This equation couples the effects of fluid viscosity (μ), fracture height (H), and leak-off coefficient (C_L), and is expressed in Equation (1) as:

$$\frac{E'}{128\mu H} \frac{\partial^2}{\partial x^2} (w_0^2) = \frac{8C_L}{\pi\sqrt{t-t_0}} + \frac{\partial w_0}{\partial t} \quad (1)$$

where E' is the effective elastic modulus, μ is fluid viscosity, H is the fracture height, C_L is the leak-off coefficient, and $w_0(x, t)$ is the fracture width. For plane-strain conditions, $E' = \frac{E}{1-\nu^2}$, where E is Young's modulus and ν is Poisson's ratio. In this study, we fix $E' = 30$ GPa as a representative shale modulus to isolate the effects of μ , H , and C_L across benchmark cases (Table 1); site-specific values can be substituted without changing the formulation. The goal of the PINN is to approximate $w_0(x, t)$ such that it satisfies the above PDE, subject to the appropriate boundary and initial conditions.

3.2 | PINN Framework

The solution $w_0(x, t)$ is approximated by a neural network $w_{PINN}(x, t; \theta)$, where θ represents the trainable weights and biases of the network. The PINN framework consists of the following key components:

3.2.1 | Neural Network Architecture

The neural network used in this study is a fully connected feedforward neural network designed to approximate the fracture width

$w_0(x, t)$ based on spatial and temporal inputs. The architecture comprises the following:

- **Input Layer:** Accepts the spatial (x) and temporal (t) variables as inputs, concatenated to represent the two-dimensional domain.
- **Hidden Layers:** Four hidden layers with 256 neurons each, employing the hyperbolic tangent ($\tanh$) activation function. The $\tanh$ activation ensures smooth approximations that are suitable for continuous physical systems governed by partial differential equations.
- **Output Layer:** A single neuron that outputs the predicted fracture width $w_0(x, t)$.

This architecture was designed to capture the spatiotemporal dependencies of fracture evolution in hydraulic fracturing simulations effectively. The depth and width of the network are chosen to balance computational efficiency with the ability to model complex physical phenomena.

3.2.2 | Loss Function

The PINN training process minimizes a composite loss function that consists of physics, boundary, and total loss components, as enumerated below.

- 1 **Physics Loss ($\mathcal{L}_{physics}$)** in Equation (2) enforces compliance with Nordgren's governing equation by penalizing residual errors at collocation points:

$$\mathcal{L}_{physics} = \frac{1}{N} \sum_{i=1}^N \left[\frac{E'}{128\mu H} \frac{\partial^2}{\partial x^2} (w_{PINN}^2) - \frac{8C_L}{\pi\sqrt{t-t_0}} - \frac{\partial w_{PINN}}{\partial t} \right]^2. \quad (2)$$

- 2 **Boundary Loss ($\mathcal{L}_{boundary}$)** enforces boundary and initial conditions, as given in Equation (3):

$$\mathcal{L}_{boundary} = \frac{1}{M} \sum_{j=1}^M [w_{PINN}(x_j, t=0) - w_0(x_j, 0)]^2. \quad (3)$$

- 3 **Total Loss ($\mathcal{L}$)** combines the physics and boundary losses resulting in Equation (4) as:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{physics} + \lambda_2 \mathcal{L}_{boundary}, \quad (4)$$

where λ_1 and λ_2 are weights to balance the contributions of each loss component.

3.2.3 | Collocation Points and Optimization

The PINN evaluates the loss function at collocation points sampled from the spatial-temporal domain; in this study they are uniformly distributed across the range of interest for x

TABLE 1 | Representative fracture parameters used for analytical solutions, PINNs, and VQE benchmarking.

Physics-Informed Neural Network (PINN)				
Parameters	Optimizers	Loss	Time (Sec)	Memory (MB, Δ RSS)
$E'=30.0, \mu=5.0, H=1.0, C_L=0.0, t_0=0.0, w(x, 1)$	Adam	3.17e-09	758.95	471.75
	Adam+L-BFGS	7.06e-08	734.31	520.00
	Adam+L-BFGS+NNCG	2.67e-08	1044.65	328.05
$E'=30.0, \mu=5.0, H=1.0, C_L=0.0, t_0=377.92, w(x, 2)$	Adam	5.74e-10	746.74	470.00
	Adam+L-BFGS	3.00e-08	724.56	344.38
	Adam+L-BFGS+NNCG	8.42e-09	1056.11	793.92
$E'=30.0, \mu=8.0, H=2.0, C_L=1.0, t_0=0.0, w(x, 1)$	Adam	2.87e-09	749.41	307.60
	Adam+L-BFGS	2.89e-07	725.02	195.61
	Adam+L-BFGS+NNCG	5.11e-08	1048.76	460.33
$E'=30.0, \mu=8.0, H=2.0, C_L=1.0, t_0=0.0, w(x, 2)$	Adam	1.23e-09	753.34	346.90
	Adam+L-BFGS	1.31e-07	743.59	977.16
	Adam+L-BFGS+NNCG	5.23e-10	1454.77	1136.56
[36] Wave-PINN convergence study	Adam	—	—	—
	Adam+L-BFGS	1.12e-3	—	—
	Adam+L-BFGS+NNCG	6.13e-5	—	—
Variational Quantum Eigensolver (VQE)				
Parameters	Optimizers	Energy minimisation	Time (Sec)	Memory (MB, Δ RSS)
$E'=30.0, \mu=5.0, H=1.0, C_L=0.0, t_0=0.0, w(x, 2)$	SPSA	−0.5825+0j	1.31	0.0
	COBYLA	−0.5831+0j	0.25	0.0
	L-BFGS-B	−0.5831+0j	0.26	0.0
$E'=30.0, \mu=8.0, H=2.0, C_L=1.0, t_0=0.0, w(x, 2)$	SPSA	−0.5825+0j	1.36	0.0
	COBYLA	−0.5831+0j	0.24	0.0
	L-BFGS-B	−0.5831+0j	0.20	0.0

and t , providing broad coverage of the domain and stable residual estimates. The network parameters θ are then trained with a staged optimizer schedule using Adam, Adam + L-BFGS, and Adam + L-BFGS + NNCG, all of which are robust gradient-based methods suited to stiff PDE residuals. Automatic differentiation computes $\partial_t w$ and $\partial_{xx}(w^2)$ for the physics loss, while boundary losses enforce constraints at the edges and initial time. In practice, the collocation density, boundary weighting, and optimizer transitions strongly influence convergence and accuracy; for example, higher point density improves residual fidelity but raises cost, whereas quasi-Newton fine-tuning sharpens convergence once Adam stabilizes the solution manifold. This workflow balances exploration and refinement to produce stable fracture-width predictions.

3.3 | Forward-Inverse PINN

The forward PINN solves the governing PDE to predict the fracture width $w_0(x, t)$ across the domain. The training process involves sampling collocation points (x_i, t_i) from the domain, computing the total loss function based on the physics and boundary residuals, and iteratively updating the network parameters using gradient descent to minimize the loss. The forward

PINN is validated against analytical solutions (if available) or experimental data to ensure accuracy.

However, the inverse PINN extends the forward PINN by incorporating unknown parameters, such as C_L and μ , as trainable variables. These parameters are optimized simultaneously with the network weights to minimize a total loss function that includes both physics and data loss components. The inverse PINN framework is particularly useful for parameter estimation in scenarios where direct measurements are unavailable.

While PINNs offer a powerful framework for solving hydraulic fracturing problems, they face challenges such as high computational costs for large-scale domains, convergence issues in heterogeneous and stochastic systems, and sensitivity to hyperparameter selection (e.g., learning rates, collocation point density). These limitations motivate the exploration of quantum computing algorithms, which are discussed in subsequent sections.

4 | Quantum Computing for Hydraulic Fracturing

As hydraulic fracturing simulations become increasingly complex, traditional computational methods face significant scalabil-

ity and efficiency challenges. These challenges are particularly pronounced in high-dimensional problems, stochastic reservoir conditions, and multi-fracture systems. Quantum computing offers a complementary approach that may help address some of these limitations, leveraging quantum mechanical principles to solve optimization and simulation problems efficiently. In this study, the Variational Quantum Eigensolver (VQE) is proposed as the primary quantum algorithm to advance hydraulic fracturing simulations. The VQE, a hybrid classical-quantum algorithm, is well suited to optimization and eigenvalue subproblems relevant to this formulation.

4.1 | Variational Quantum Eigensolver (VQE)

VQE is a quantum-classical hybrid algorithm designed to solve optimization problems by approximating the ground state of a Hamiltonian, which represents the energy of a system. It combines the quantum computer's ability to prepare and measure quantum states with a classical optimizer that minimizes an objective function-based on these measurements.

In the context of hydraulic fracturing, the Nordgren equation governing fracture dynamics can be reformulated into an optimization problem, where the fracture width $w_0(x, t)$ is represented as the state variable to minimize the residual of the PDE in Equation (5):

$$\mathcal{R}(x, t) = \frac{E'}{128\mu H} \frac{\partial^2}{\partial x^2} (w_0^2) - \frac{8C_L}{\pi\sqrt{t-t_0}} - \frac{\partial w_0}{\partial t}. \quad (5)$$

The objective of the VQE is to minimize the residual over the domain by iteratively improving the parameterized quantum state until it closely approximates the solution $w_0(x, t)$.

To integrate the Variational Quantum Eigensolver (VQE) into hydraulic fracturing simulations, we need to reformulate the Nordgren equation into a mathematical representation suitable for quantum computation. Below is the detailed mathematical framework behind the VQE integration:

4.1.1 | Reformulating the Nordgren Equation

The Nordgren equation is repeated in Equation (6) from Equation (1):

$$\frac{E'}{128\mu H} \frac{\partial^2}{\partial x^2} (w_0^2) = \frac{8C_L}{\pi\sqrt{t-t_0}} + \frac{\partial w_0}{\partial t}, \quad (6)$$

This equation can be discretized over the domain (x, t) using finite-difference or spectral methods. The discretization transforms the PDE into a system of linear or nonlinear equations that can be solved computationally.

4.1.2 | Discretization of Spatial and Temporal Derivatives

1 Discretize Spatial Domain (x):

Let $x \in [0, L]$ be divided into N_x grid points. The second derivative w_0^2 is approximated using finite differences in Equation (7):

$$\frac{\partial^2}{\partial x^2} (w_0^2) \approx \frac{w_0^2(x_{i+1}, t) - 2w_0^2(x_i, t) + w_0^2(x_{i-1}, t)}{\Delta x^2}, \quad (7)$$

where $\Delta x = \frac{L}{N_x - 1}$.

2 Discretize Time Domain (t) in Equation (8):

Let $t \in [t_0, T]$ be divided into N_t grid points. The temporal derivative is approximated as:

$$\frac{\partial w_0}{\partial t} \approx \frac{w_0(x_i, t_{n+1}) - w_0(x_i, t_n)}{\Delta t}, \quad (8)$$

where $\Delta t = \frac{T-t_0}{N_t - 1}$.

4.1.3 | Resulting Discretized System

Combining these discretizations from Equations (7) and (8) rewrites the Nordgren equation in Equation (10) as:

$$\frac{E'}{128\mu H} \cdot \frac{w_0^2(x_{i+1}, t_n) - 2w_0^2(x_i, t_n) + w_0^2(x_{i-1}, t_n)}{\Delta x^2} \quad (9)$$

$$= \frac{8C_L}{\pi\sqrt{t_n - t_0}} + \frac{w_0(x_i, t_{n+1}) - w_0(x_i, t_n)}{\Delta t}. \quad (10)$$

Rearranging into matrix form for all x_i and t_n , we get Equation (11):

$$\mathbf{Aw} = \mathbf{b}, \quad (11)$$

where $\mathbf{w}$ is the vector of fracture widths w_0 at all grid points, $\mathbf{A}$ encodes the spatial and temporal derivatives and coefficients, and $\mathbf{b}$ represents the source term $\frac{8C_L}{\pi\sqrt{t-t_0}}$.

4.2 | Encoding the System for VQE

In VQE, the linear system $\mathbf{Aw} = \mathbf{b}$ is encoded into a Hamiltonian, where the solution $\mathbf{w}$ corresponds to the eigenstate with the lowest eigenvalue (ground state).

4.2.1 | Hamiltonian Encoding, Quantum Mapping, and VQE Objective

The system $\mathbf{Aw} = \mathbf{b}$ is converted into a least-squares optimization problem, shown in Equation (12) as:

$$\min_{\mathbf{w}} \|\mathbf{Aw} - \mathbf{b}\|^2, \quad (12)$$

which is equivalent to finding the ground state of the Hamiltonian in Equation (13):

$$H = \mathbf{A}^\dagger \mathbf{A} - \mathbf{A}^\dagger \mathbf{b} - \mathbf{b}^\dagger \mathbf{A} + \mathbf{b}^\dagger \mathbf{b}. \quad (13)$$

The matrix $\mathbf{A}$ and vector $\mathbf{b}$ are normalized to keep coefficients within hardware-friendly ranges and to improve conditioning. The Hamiltonian is then expressed as a weighted sum of tensor products of Pauli operators (I, X, Y, Z) using mappings such as Jordan–Wigner or Bravyi–Kitaev; Equation (14) captures the resulting operator form:

$$H = \sum_{i,j} h_{ij} Z_i Z_j + \sum_i c_i Z_i + \text{constant}. \quad (14)$$

The variational ansatz is a parameterized quantum circuit representing candidate solutions $\mathbf{w}$, defined in Equation (15):

$$|\psi(\theta)\rangle = U(\theta)|0\rangle, \quad (15)$$

where $U(\theta)$ is built from layers of single-qubit rotations and entangling gates. The ansatz depth controls expressivity: deeper circuits can represent richer states but incur higher noise and measurement costs, so a balanced depth is selected. The VQE objective is the expectation value of the Hamiltonian,

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle, \quad (16)$$

which is estimated from repeated measurements. A classical optimizer adjusts θ to minimize $E(\theta)$, aiming to recover the ground state that corresponds to the optimal $\mathbf{w}$. Practical performance depends on initialization, optimizer choice, and shot budget; these factors govern convergence speed and robustness in noisy settings.

5 | Results and Discussion

5.1 | Nordgren PINN Development and Optimization

The Nordgren PINN workflow in Algorithm 1 encodes physics into the neural network’s training process. First, the neural network takes spatial (x) and temporal (t) coordinates as inputs, outputting the predicted fracture width $w(x, t)$. The network architecture includes fully connected layers with non-linear activation functions (e.g., tanh) to capture complex relationships.

Two loss functions govern optimization: the physical residual loss enforces Nordgren’s equation, while the boundary condition loss ensures that $w(x, t)$ aligns with known values at the boundaries. The total loss is a weighted combination of these, with a hyperparameter (λ_{bc}) balancing their contributions.

Training begins with the Adam optimizer for efficient parameter updates, followed by L-BFGS and Newton-CG optimizers for fine-tuning to achieve better convergence. Finally, the trained PINN predicts $w(x, t)$ for test points, providing a physics-constrained approximation of fracture dynamics. This physics-informed approach ensures compliance with governing equations while leveraging neural networks’ ability to generalize.

ALGORITHM 1 | Nordgren PINN development and optimization.

- 1: **Inputs:** E', μ, H, C_L, t_0 , collocation set D_f , boundary/initial set D_b , weight λ_{bc}
- 2: **Output:** Trained network $w_\theta(x, t)$
- 3: **Initialize Network:**
Fully connected layers with tanh activation; raw output $\tilde{w}_\theta(x, t)$
- 4: **Non-negativity (optional):**
 $w_\theta(x, t) = \text{softplus}(\tilde{w}_\theta)$ or add a penalty for $w_\theta < 0$
- 5: **Define PDE Residual:**
$$R(x, t) = \frac{E'}{128\mu H} \frac{\partial^2}{\partial x^2} (w_\theta^2) - \frac{8C_L}{\pi\sqrt{t-t_0}} - \frac{\partial w_\theta}{\partial t}$$
- 6: **Losses:**
$$L_f = \text{MSE}_{(x,t) \in D_f} [R(x, t)]$$

$$L_{bc} = \text{MSE}_{(x,t) \in D_b} [w_\theta(x, t) - w_{bc}(x, t)]$$

If using penalty: $L_{pos} = \text{MSE}(\min(w_\theta, 0), 0)$
$$L = L_f + \lambda_{bc} L_{bc} + \lambda_{pos} L_{pos}$$
- 7: **Training:**
Sample/update D_f, D_b
Optimize L with Adam, then L-BFGS and Newton-CG
- 8: **Evaluate:**
Predict $w_\theta(x, t)$ on a test grid

5.1.1 | Neural Convergence and Fracture Evolution Across Optimizers

This comparative analysis in Figure 1 evaluates the performance of three optimization techniques—Adam, Adam + L-BFGS, and Adam + L-BFGS + NNCG—for training a PINN model on Nordgren’s hydraulic fracturing problem. The first row of training loss curves shows that Adam converges smoothly but at lower final precision. Adding L-BFGS improves convergence, reducing the loss significantly, while the combination of Adam, L-BFGS, and NNCG further accelerates convergence with higher stability, albeit with an initially higher loss due to adaptive reweighting. The second row compares PINN-predicted fracture width with the analytical solution at $t = 1.0$. Across all optimizers, the PINN struggles to match the analytical quadratic growth trend, significantly underestimating the fracture width. This discrepancy suggests either insufficient network capacity or unresolved nonlinearities in training, though Adam + L-BFGS + NNCG shows the closest fit. The third row presents contour maps of $w_0(x, t)$ over the spatiotemporal domain. While smoothness improves from Adam to Adam + L-BFGS + NNCG, the fracture width predictions remain largely negative or close to zero, deviating from expected physical behavior. These results highlight the combined optimizers’ ability to enhance PINN convergence but also underscore persistent challenges in accurately capturing fracture dynamics, possibly due to model limitations or hyperparameter settings.

The comparison evaluates Adam, Adam + L-BFGS, and Adam + L-BFGS + NNCG optimizers for training a PINN on Nordgren’s equation. In Figure 2, the training loss curves reveal rapid

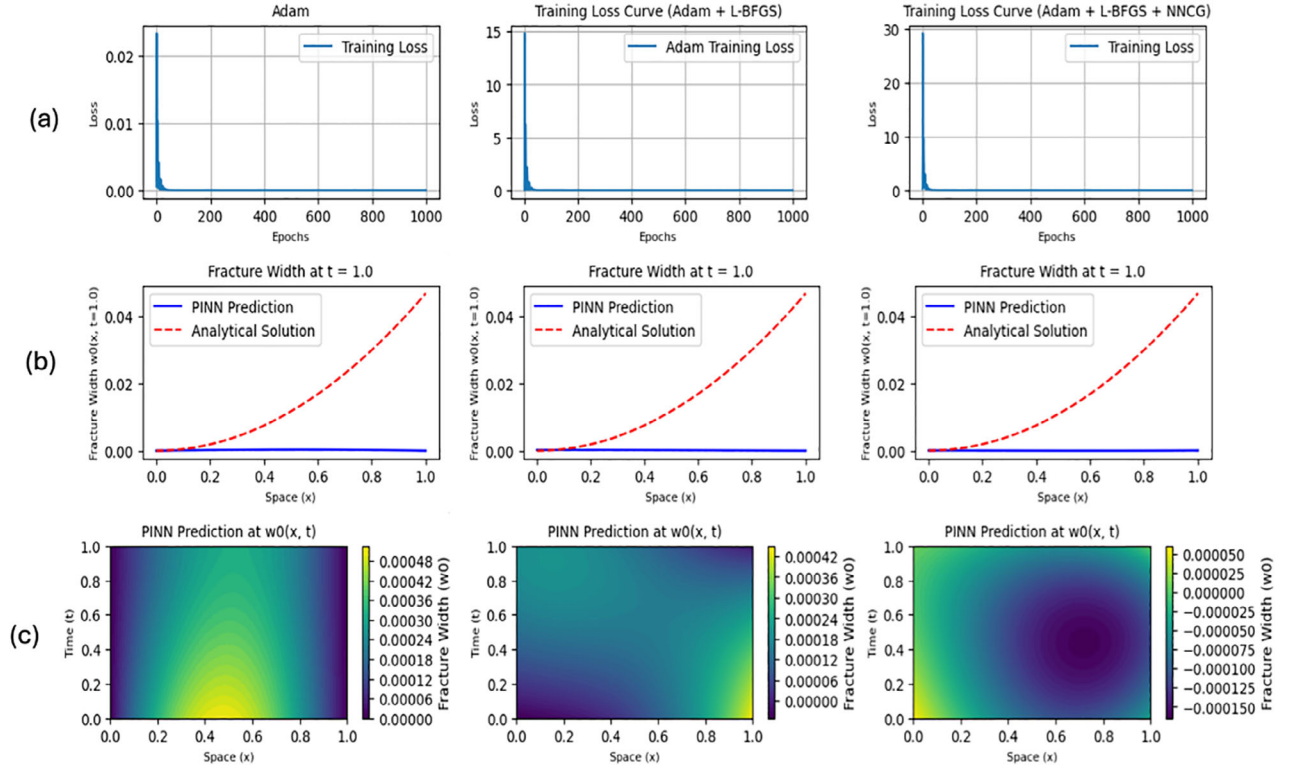


FIGURE 1 | Fracture parameters: $E' = 30.0$, $\mu = 5.0$, $H = 1.0$, $C_L = 0.0$, $t_0 = 0.0$. (a) Loss curves showing the training performance of three distinct optimizers. (b) Comparison of PINN-predicted fracture width evolution with the analytical solution at $t = 1.0$. (c) Spatiotemporal dynamics of fracture width $w_0(x, t)$ as predicted by PINN.

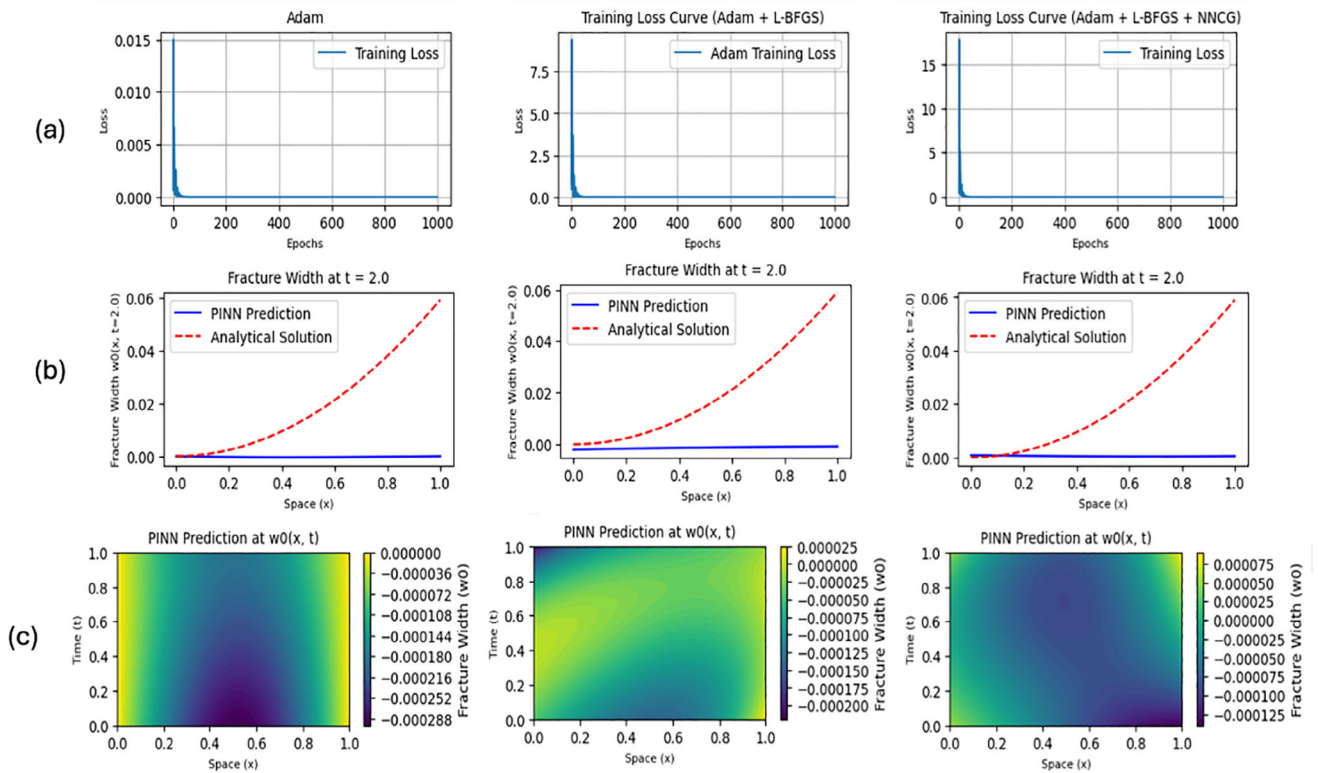


FIGURE 2 | Fracture parameters: $E' = 30.0$, $\mu = 5.0$, $H = 1.0$, $C_L = 0.0$, $t_0 = 0.0$. (a) Loss curves showing the training performance of three distinct optimizers. (b) Comparison of PINN-predicted fracture width evolution with the analytical solution at $t = 2.0$. (c) Spatiotemporal dynamics of fracture width $w_0(x, t)$ as predicted by PINN.

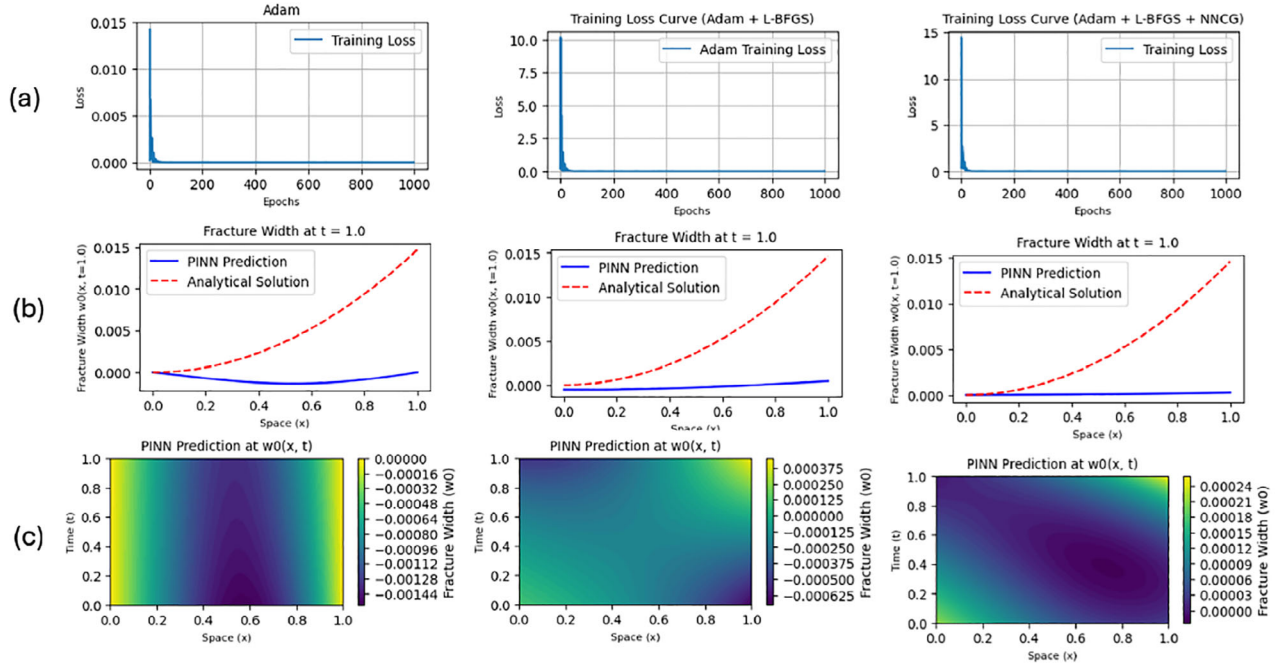


FIGURE 3 | Fracture parameters: $E' = 30.0$, $\mu = 8.0$, $H = 2.0$, $C_L = 1.0$, $t_0 = 0.0$. (a) Loss curves showing the training performance of three distinct optimizers. (b) Comparison of PINN-predicted fracture width evolution with the analytical solution at $t = 1.0$. (c) Spatiotemporal dynamics of fracture width $w_0(x, t)$ as predicted by PINN.

convergence for all methods, with losses stabilizing at ≈ 0.005 (Adam), ≈ 0.002 (Adam + L-BFGS), and ≈ 0.0015 (Adam + L-BFGS + NNCG). The fracture width predictions at $t = 2.0$ show that all methods underestimate the analytical solution's quadratic growth, with Adam + L-BFGS + NNCG providing the closest match at ≈ 0.0025 . The spatiotemporal contour maps illustrate smooth predictions, but negative values dominate, peaking at ≈ 0.00025 for Adam + L-BFGS + NNCG. Despite improvement, accuracy gaps remain.

Figure 3 compares the performance of three optimization techniques (Adam, Adam + L-BFGS, and Adam + L-BFGS + NNCG) for training PINNs on Nordgren's equation. The training loss curves show rapid convergence, with Adam stabilizing around Loss ≈ 0.002 , Adam + L-BFGS around Loss ≈ 0.001 , and Adam + L-BFGS + NNCG achieving the lowest loss at Loss ≈ 0.0005 . Fracture width predictions at $t = 1.0$ show underestimation relative to the analytical solution for all methods, with Adam + L-BFGS + NNCG giving the closest match. Spatiotemporal contour maps show localized positive values, but the overall trend still underestimates the expected growth of $w_0(x, t)$ over time. These results indicate that combined optimizers improve convergence but do not fully resolve PINN accuracy limitations for this case.

Figure 4 evaluates the performance of three optimization strategies—Adam, Adam + L-BFGS, and Adam + L-BFGS + NNCG—for predicting fracture width $w_0(x, t)$ in hydraulic fracturing. The training loss curves indicate that all methods rapidly reduce loss, with Adam converging to ≈ 0.002 , Adam + L-BFGS to ≈ 0.001 , and Adam + L-BFGS + NNCG to ≈ 0.0005 . At $t = 2.0$, PINN predictions still underestimate the analytical solution, with only limited improvement across methods. The spatiotemporal contour maps illustrate the evolution of $w_0(x, t)$; regions

in green/yellow indicate relatively higher predicted values but remain below expected results. These findings highlight persistent challenges in capturing accurate fracture-width dynamics despite improved optimization.

The systematic underestimation of crack width is likely driven by a combination of factors: (i) scale imbalance in the PDE residual due to the w^2 term and the lack of nondimensionalization, which can bias optimization toward near-zero solutions; (ii) sparse boundary supervision (only two boundary points) and fixed loss weighting, allowing residual minimization to dominate; (iii) the absence of an explicit non-negativity constraint on w , which permits negative predictions; and (iv) collocation density and optimizer convergence to local minima in a stiff, nonlinear residual landscape. Remedies include nondimensionalization, increasing boundary/collocation points, adaptive loss weighting or residual-based refinement, and enforcing $w \geq 0$ (e.g., softplus output or penalty).

5.2 | Nordgren VQE Development and Optimization

This part of the computation employs quantum variational algorithms, as demonstrated in Algorithm 2, to approximate fracture widths governed by Nordgren's equation. Initially, physical parameters such as E' (effective elastic modulus), μ (fluid viscosity), and H (fracture height) are defined. These constants are used to discretize Nordgren's equation into a Hamiltonian matrix, representing the system's quantum state.

The core of the approach is the variational ansatz, a parameterized quantum circuit constructed using a specified number of qubits. Each qubit undergoes parameterized single-qubit

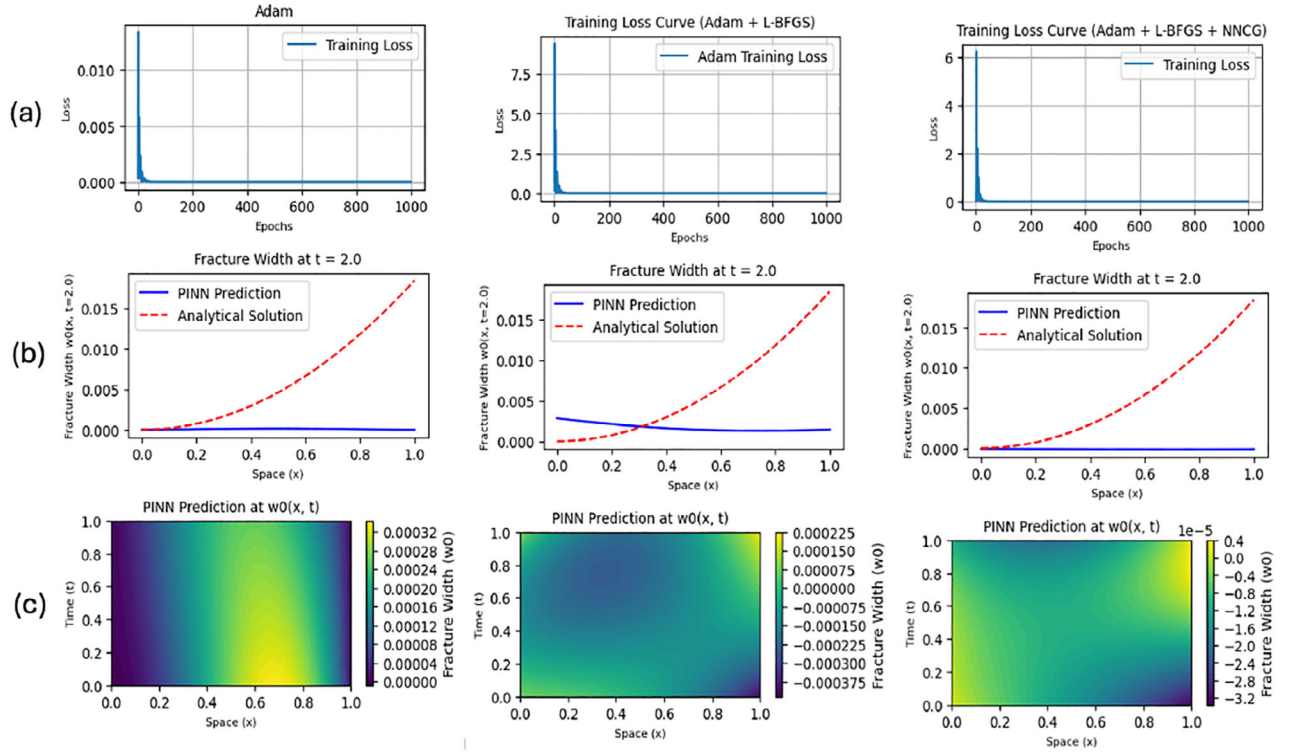


FIGURE 4 | Fracture parameters: $E' = 30.0$, $\mu = 8.0$, $H = 2.0$, $C_L = 1.0$, $t_0 = 0.0$. (a) Loss curves showing the training performance of three distinct optimizers. (b) Comparison of PINN-predicted fracture width evolution with the analytical solution at $t = 2.0$. (c) Spatiotemporal dynamics of fracture width $w_0(x, t)$ as predicted by PINN.

rotations and entanglement operations, forming the quantum circuit's backbone. This ansatz captures the system's trial wavefunction, essential for solving the eigenvalue problem.

Using optimizers like SPSA, COBYLA, or L-BFGS-B, the VQE algorithm iteratively minimizes the Hamiltonian's energy expectation value, solving for the lowest eigenvalue. A callback function tracks energy convergence during this process, which helps monitor optimization stability.

Post-optimization, the eigenvalue is mapped to the fracture width using physics-based scaling factors. VQE predictions are generated across a spatiotemporal grid, enabling a comparison with analytical solutions. The results include energy convergence plots, spatial profiles of fracture width, and contour maps showing the evolution of width over time and space. This framework provides a practical interface between quantum computation and physics-based modeling.

5.2.1 | Quantum Circuit, State, and Measurement

After encoding Nordgren's fracture model into the Variational Quantum Eigensolver, fracture-width optimization began by constructing a quantum circuit. Figure 5a shows the parameterized VQE ansatz, which uses single-qubit rotations (R_x , R_y , R_z) and entanglement via controlled-X gates. This layout parameterizes the qubits for the Nordgren-based objective while capturing inter-qubit correlations. The rotation gates were defined with six parameters, $\theta[0]$ to $\theta[5]$, enabling tunable exploration of the energy landscape. By entangling qubits, the circuit can

represent nontrivial wavefunctions needed for PDE-constrained optimization tasks.

The resulting qubit states, visualized on the Bloch spheres in Figure 5b, show the optimized state orientations for each qubit. The arrows represent quantum states projected in Bloch coordinates. These visualizations suggest that the optimized parameter settings produce distinct quantum states. The density matrices in Figure 5c are shown as 3D bar plots of real and imaginary components in the computational basis. The compact distribution indicates concentration around a target state. The Wigner function in Figure 5d represents the phase-space distribution and highlights interference patterns characteristic of quantum systems. The histogram in Figure 5e reports measurement outcomes after simulation, where a dominant basis state indicates convergence toward a low-energy configuration. Overall, these diagnostics are consistent with stable VQE optimization for the tested setup.

Nordgren's model is then evaluated using the provided pseudocode to estimate fracture-width evolution under the parameter sets in Table 1 over time. This study focuses on three optimizers (SPSA, COBYLA, and L-BFGS-B) to compare convergence behavior and noise sensitivity.

5.2.2 | Energy Convergence and Fracture Evolution Across Optimizers

For the parameter set with shale Young's modulus fixed at 30 GPa, fluid viscosity $\mu = 5.0 \text{ Pa} \cdot \text{s}$, fracture height $H = 1 \text{ m}$, and zero

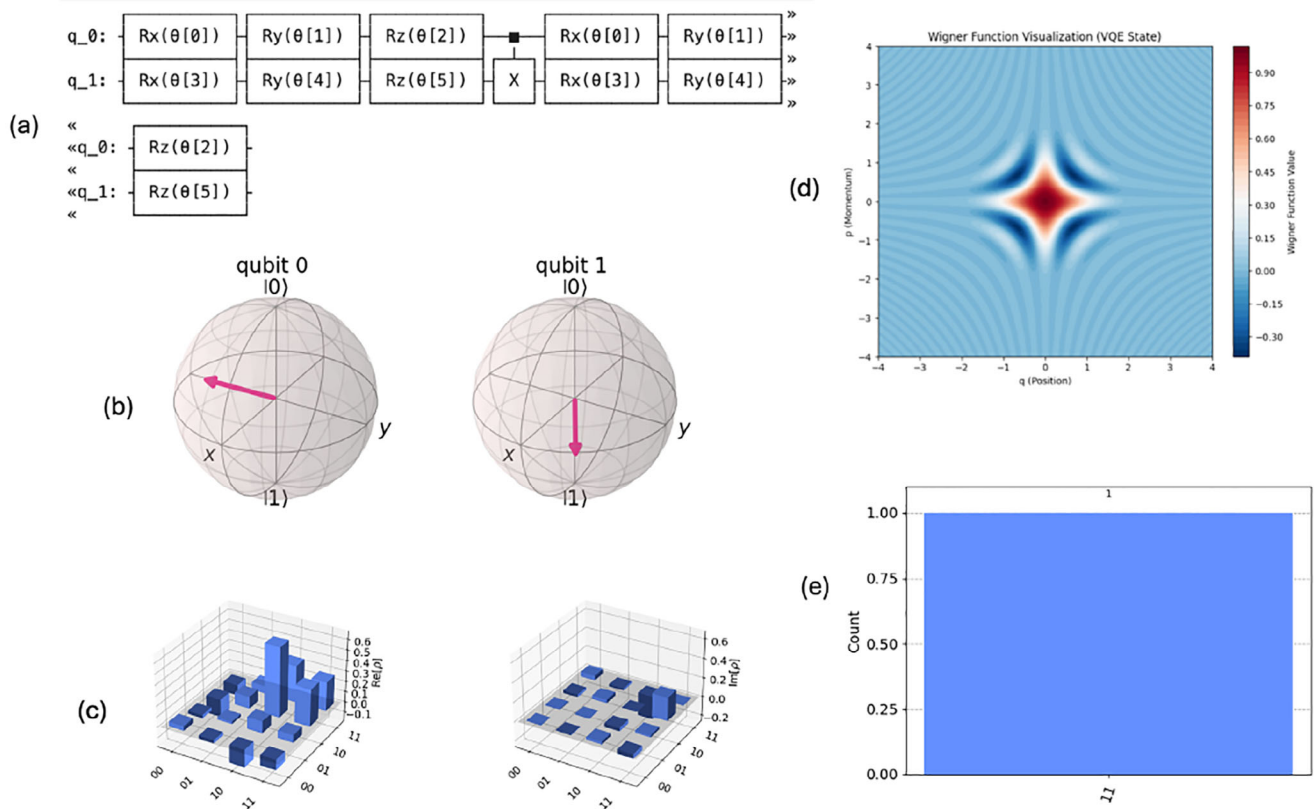


FIGURE 5 | Quantum computing simulator framework for optimizing VQE to solve Nordgren's fracture model. (a) Quantum circuit architecture. (b) Single-qubit state visualizations. (c) Statevector representations of the VQE ansatz. (d) Wigner function plots illustrating quantum state characteristics. (e) Measurement results and probability distributions.

leak-off coefficient, the energy-convergence plots compare SPSA, COBYLA, and L-BFGS-B. SPSA shows gradual stabilization over 1000 iterations with stochastic fluctuations, consistent with its gradient-free stochastic nature. COBYLA converges more rapidly (within about 200 iterations) and remains stable thereafter. L-BFGS-B reaches convergence within roughly 100 iterations, as expected for smoother objective landscapes. Figure 6a therefore shows a clear trade-off between computational cost and convergence behavior across optimizers.

The fracture-width predictions at $t = 2.0$ in Figure 6b illustrate spatial variations captured by VQE relative to the analytical solution. While the analytical profile is smooth and parabolic, VQE predictions show oscillations and an offset. These differences may arise from ansatz expressivity limits, optimizer behavior, or measurement noise. This indicates the need for ansatz refinement and additional noise-mitigation strategies to improve fidelity. The contour maps of $w_0(x, t)$ in Figure 6c show broadly similar spatial-temporal patterns across optimizers, with COBYLA and L-BFGS-B appearing smoother than SPSA. Overall, the results suggest that VQE captures qualitative trends but still exhibits quantitative errors that should be reduced in future work.

In a second test case, fluid viscosity was increased to $\mu = 8.0$ Pa · s, while fracture height and leak-off coefficient were set to 2 and 1, respectively. Figure 7a shows energy convergence for SPSA, COBYLA, and L-BFGS-B. SPSA required the highest iteration count (1000) and retained stochastic fluctuations, which is

consistent with its gradient-free stochastic update rule. COBYLA converged within about 350 iterations with a smoother descent. L-BFGS-B converged fastest, stabilizing within about 100 iterations. All three optimizers approached an energy near -0.583, indicating similar minima for this setup.

Figure 7b compares fracture-width predictions at $t = 2.0$ for SPSA, COBYLA, and L-BFGS-B. The red dashed line is the analytical solution (smooth parabola), whereas the VQE predictions (blue lines) remain oscillatory. These discrepancies are consistent with finite ansatz capacity and noise in the variational procedure. L-BFGS-B shows slightly lower variance than the other optimizers in this case. Figure 7c presents spatiotemporal contour maps of VQE-predicted fracture width across optimizers. The global trends are similar across methods, while local intensity and smoothness vary slightly, reflecting differences in optimization quality. Overall, VQE captures broad trends but still shows non-negligible local deviations from the analytical profile.

5.3 | Nordgren Fracture Analysis

The Nordgren fracture analysis employs two distinct computational architectures, the Physics-Informed Neural Network (PINN) and the Variational Quantum Eigensolver (VQE), to predict the evolution of fracture width under varying physical parameters. These architectures differ fundamentally in their computational frameworks, optimization strategies, and effi-

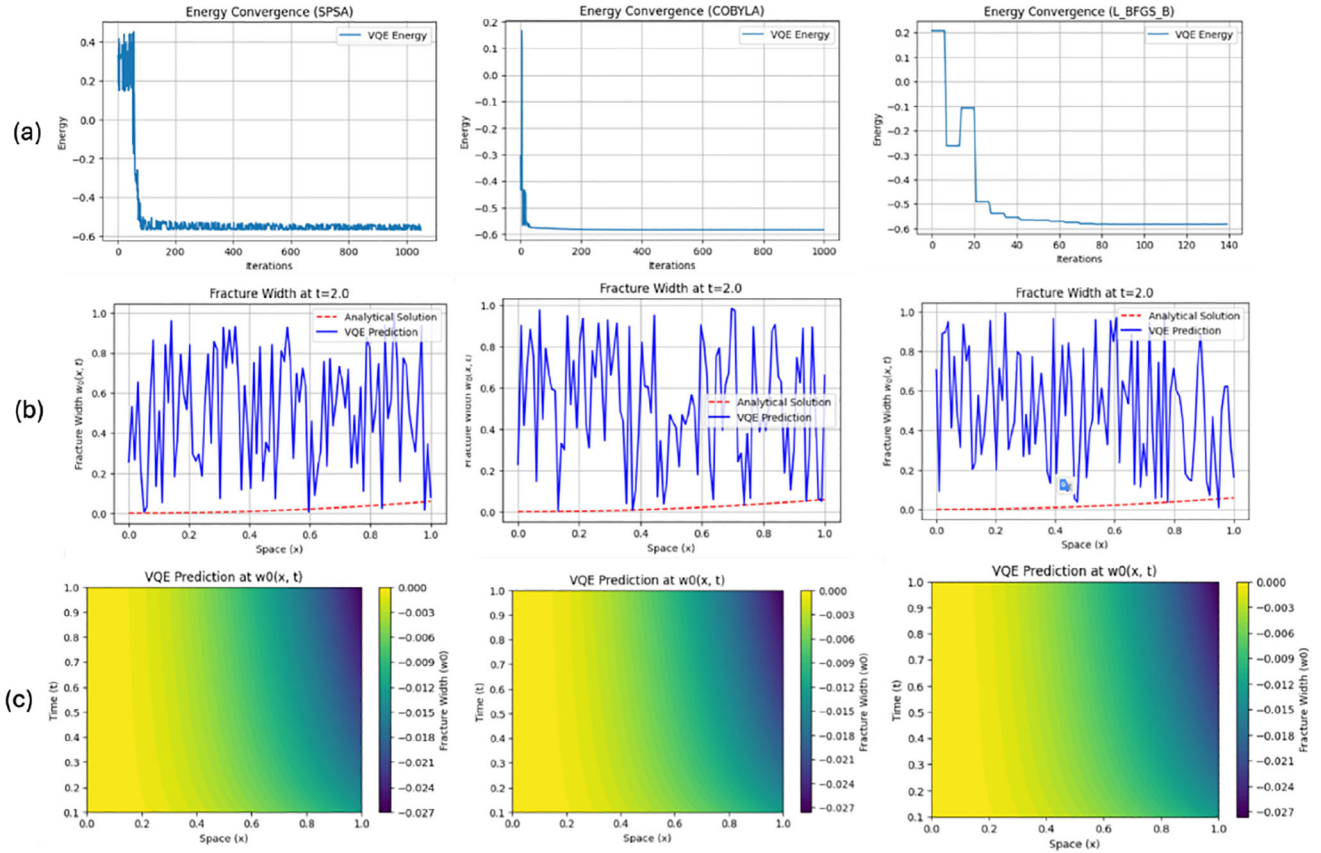


FIGURE 6 | Fracture parameters: $E' = 30.0$, $\mu = 5.0$, $H = 1.0$, $C_L = 0.0$, $t_0 = 0.0$. (a) Loss curves showing the training performance of three distinct optimizers. (b) Comparison of VQE-predicted fracture width evolution with the analytical solution at $t = 2.0$. (c) Spatiotemporal dynamics of fracture width $w_0(x, t)$ as predicted by VQE.

ciency in delivering solutions of different accuracy–cost profiles, as shown in Table 1. The parameter sets in Table 1 are representative benchmarks used for method comparison rather than site-specific field measurements.

Note: Memory values report the incremental resident set size (Δ RSS) of the Python process measured with `memory_profiler` in Google Colab; GPU memory is not included. The Adam+L-BFGS+NNCG row for $t_0 = 377.92$ was corrected from a time/memory swap in the earlier draft.

5.3.1 | PINN Analysis

PINNs leverage the physical principles embedded in partial differential equations (PDEs) and boundary conditions to model fracture width evolution. The PINN framework uses a fully connected neural network architecture with tanh activation functions to approximate the solution of the Nordgren equation. It minimizes a combined loss function comprising the residual loss from the PDE and the boundary loss, with a regularization term λ_{bc} to enforce boundary conditions. The training of PINNs, as seen in Table 1, is computationally intensive, with Adam optimization followed by fine-tuning using L-BFGS and Newton-CG contributing significantly to runtime and memory use. For example, the Adam+L-BFGS+NNCG configuration required over 1000 s and up to 1136 MB of memory for some parameter sets.

Despite higher computational requirements, PINN predictions can exhibit smooth spatiotemporal evolution of fracture width, especially when boundary constraints are enforced effectively. The loss minimization of PINNs is sensitive to fracture parameters such as viscosity μ , height H , and Young's modulus E' , which influence residual scaling and optimization behavior. Table 1 indicates case-dependent loss variation as μ changes from 5.0 to 8.0; this should be interpreted as an optimization effect under the selected setup rather than a universal improvement in predictive accuracy.

5.3.2 | VQE Analysis

In contrast, the VQE architecture, on the other hand, relies on quantum circuit-based optimization to solve for the minimum eigenvalue of a discretized Hamiltonian, which is mapped to the fracture width evolution. The ansatz design, incorporating parameterized single-qubit rotations and entanglement layers, forms the foundation for representing the quantum state of the system. Unlike PINNs, VQE optimizes directly over quantum states using classical optimizers such as SPSA, COBYLA, and L-BFGS-B.

In the tested benchmark cases, VQE required substantially less wall-clock time and memory than PINNs. For the same fracture parameters, VQE achieves convergence in under 2 s with neg-

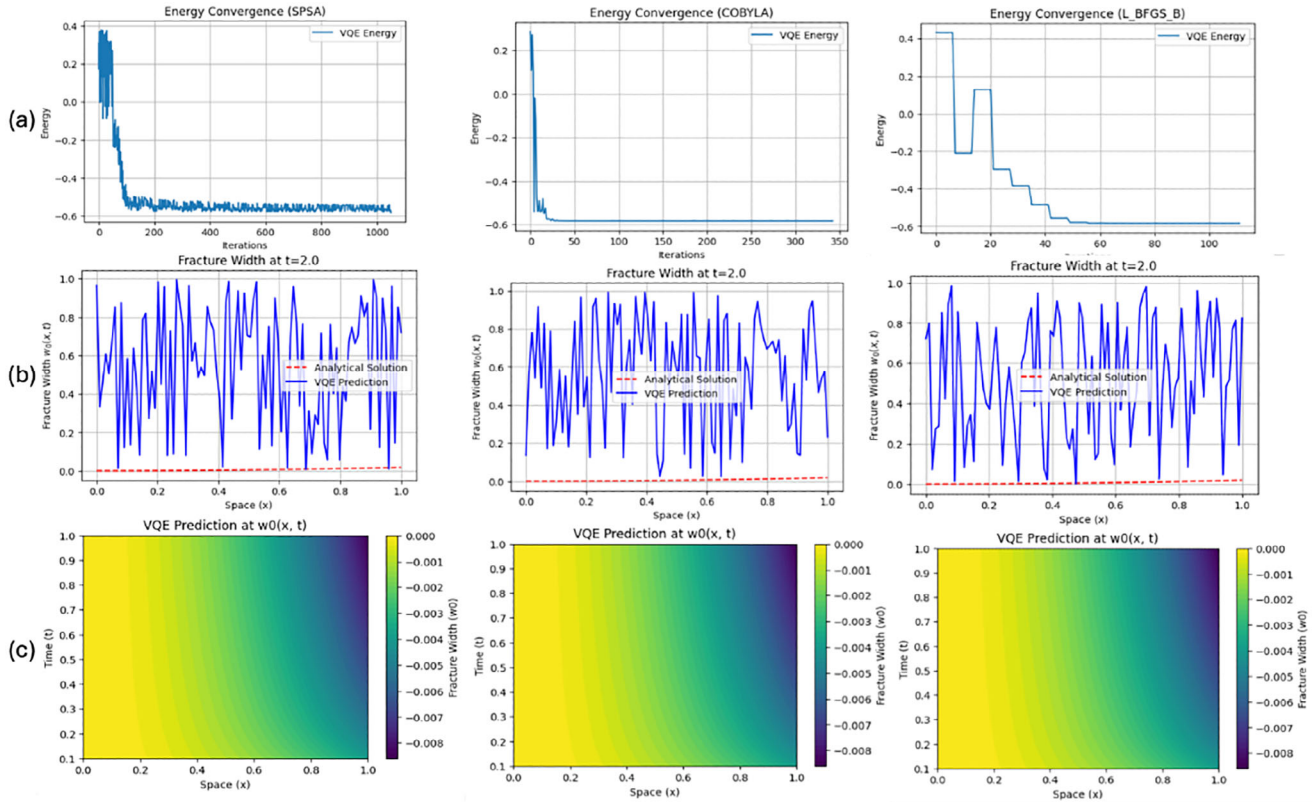


FIGURE 7 | Fracture parameters: $E' = 30.0$, $\mu = 8.0$, $H = 2.0$, $C_L = 1.0$, $t_0 = 0.0$. (a) Loss curves showing the training performance of three distinct optimizers. (b) Comparison of VQE-predicted fracture width evolution with the analytical solution at $t = 2.0$. (c) Spatiotemporal dynamics of fracture width $w_0(x, t)$ as predicted by VQE.

ligible memory requirements, as shown in Table 1. The energy minimization results, consistently converging to approximately -0.583 , indicate stable optimizer behavior for these cases. While VQE sacrifices some of the interpretability and flexibility of PINNs (e.g., in explicitly encoding boundary conditions), its faster convergence and lower computational overhead make it suitable for quick predictions in high-dimensional parameter spaces.

5.3.3 | Disparities

The fundamental difference between PINNs and VQE lies in their approaches to optimization and how they handle physical constraints. PINNs inherently encode the Nordgren equation and boundary conditions into the loss function, which generally promotes physically consistent solutions. However, this tight coupling with physics leads to higher computational costs and potential challenges in convergence for complex parameter sets. In contrast, VQE decouples the optimization process from explicit boundary constraints, focusing purely on eigenvalue minimization. While this results in faster computations, the accuracy of fracture-width predictions may depend heavily on the quality of the Hamiltonian discretization and the ansatz design.

Overall, both architectures demonstrate strengths and weaknesses in predicting fracture width evolution. PINNs are generally advantageous for handling complex boundary conditions and enforcing physical consistency, while VQE offers a faster, memory-efficient alternative for problems with well-defined

Hamiltonians. The choice between these methods depends on problem requirements, with PINNs being more suitable for detailed fracture modeling and VQE favored for rapid quantum-assisted computations.

6 | Validation With Implicit Scheme

While the Physics-Informed Neural Network (PINN) and Variational Quantum Eigensolver (VQE) provide advanced computational paradigms for solving Nordgren's hydraulic fracturing model, their reliability requires validation against established numerical techniques. To benchmark the accuracy of these methods, an implicit numerical scheme was employed as a reference solver for the governing partial differential equations (PDEs) of the Nordgren model.

The implicit numerical scheme is a robust and widely used method for solving time-dependent PDEs due to its inherent stability, especially for stiff equations. The scheme discretizes the governing equations spatially using finite difference approximations and employs a time-stepping procedure to iteratively compute fracture-width evolution, $w_0(x, t)$. The implicit formulation maintains stability for larger time steps, making it suitable for validating solutions generated by PINNs and VQE.

The spatial domain $x \in [0, 1]$ was discretized into 100 points, and the temporal domain $t \in [0, 2]$ was divided into 200 steps. The governing Nordgren equation was rewritten in its discretized

1: **Inputs:** E' , μ , H , C_L , t_0 , grid (x_i, t_k) , ansatz depth, optimizer

2: **Output:** Approximate $w(x, t)$ or ground-state energy

3: **Discretize Nordgren's PDE:**

$$R = \frac{E'}{128\mu H} \frac{\partial^2}{\partial x^2} (w^2) - \frac{8C_L}{\pi \sqrt{t-t_0}} - \frac{\partial w}{\partial t}$$

Assemble objective (e.g., $\|R\|^2$) and map to a Hamiltonian $H = \sum_i c_i P_i$

4: **Define Ansatz Circuit:**

Choose n qubits and parameters $\theta = \{\theta_0, \dots, \theta_m\}$

Apply parameterized single-qubit rotations and entangling gates; repeat layers as needed

5: **Set Up VQE:**

Choose optimizer (e.g., SPSA, COBYLA, or L-BFGS)

Minimize $E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$ with callback tracking

6: **Post-process:**

3: Use θ^* (or $\lambda_{\min}$) with physics-based scaling to recover $w(x, t)$

7: **Analytical similarity (benchmark, zero leak-off):**

$w_{\text{sim}}(x, t) = t^{1/3} f(\xi)$ with $\xi = x/t^{1/3}$, where f satisfies the Nordgren similarity ODE [35]

(Optional simplified benchmark)

$$w_{\text{bench}}(x, t) = \frac{E'}{128\mu H} x^2 t^{1/3}$$

Compare $w_{\text{VQE}}(x, t)$ with analytical/benchmark solutions

Plot energy convergence and contour maps of $w(x, t)$

form as:

$$\frac{w^{n+1} - w^n}{\Delta t} = \frac{E'}{128\mu H} \frac{\partial^2 w^{n+1}}{\partial x^2} \quad (17)$$

where w^n and w^{n+1} represent the fracture widths at the current and next time steps, respectively. The system of equations was solved iteratively using the Thomas algorithm, a direct solver for tridiagonal matrices.

The implicit-scheme results were benchmarked against predictions from the Physics-Informed Neural Network (PINN) and Variational Quantum Eigensolver (VQE) models for fracture parameters $E' = 30.0$, $\mu = 5.0$, $H = 1.0$, and $t_0 = 0.0$. The implicit scheme provided a stable reference depiction of $w_0(x, t)$ across the spatiotemporal domain (see Figure 8).

Comparative analysis showed that VQE predictions align well with the implicit scheme for the tested cases, while PINN results exhibit systematic underestimation for some parameter settings. Errors are therefore parameter- and region-dependent, with larger deviations in zones of rapid spatial or temporal variation. These deviations highlight the sensitivity of PINNs to hyperparameter settings (architecture, optimizer configuration, loss weighting) and the need for improved VQE ansatz depth and parameter optimization to achieve higher fidelity.

This validation study supports the applicability of both PINN and VQE approaches to hydraulic-fracturing simulation, while also identifying scenario-dependent limitations. Future work could explore adaptive techniques, such as residual-based refinement for PINNs or hybrid quantum-classical circuits for VQE, to further minimize deviations and enhance accuracy.

6.1 | Practical Application

The Nordgren fracture analysis, supported by implicit-scheme validation, has practical relevance for hydraulic-fracturing design and subsurface reservoir management. The implicit scheme serves as a reference baseline for checking PINN and VQE outputs. In practical use, PINN-based modeling is appropriate for scenarios that require stronger physics consistency, such as evaluating injection-rate schedules and fracture propagation behavior in heterogeneous formations. By contrast, the VQE workflow in this study (tested on a simulator backend) is better interpreted as a rapid-screening tool for candidate scenarios rather than a standalone field-deployment replacement.

The implicit benchmark also highlights where each method agrees or deviates, which guides targeted model refinement. A practical workflow is to use PINN (and/or the implicit solver) for higher-fidelity validation and use VQE-style optimization for fast parameter sweeps and preliminary ranking of operating conditions. Engineers can then combine these outputs for better-informed treatment design, risk screening, and iterative updating during planning.

7 | Conclusion

This study evaluates the combined use of implicit schemes, Physics-Informed Neural Networks (PINNs), and Variational Quantum Eigensolvers (VQEs) for modeling fracture-width evolution governed by Nordgren's equation. Implicit schemes serve as a reliable baseline for validating PINN and VQE predictions and checking their physical plausibility.

PINNs embed governing physical laws into the loss function, which can improve fracture-width predictions across diverse parameter configurations. However, their computational demands, including memory and runtime, highlight the need for optimization in scenarios requiring faster computations. The implicit scheme aids in pinpointing discrepancies in regions of rapid spatial or temporal fracture-width changes, guiding refinements in PINN architectures.

VQEs, on the other hand, offer rapid convergence and low memory requirements, making them promising for rapid screening applications in resource extraction and fracture monitoring. Optimizers like SPSA, COBYLA, and L-BFGS-B achieve efficient energy minimization but require further tuning to match the precision of PINNs in fracture-width prediction. The implicit scheme further validates VQE predictions, providing a consistency check against the underlying physical model.

This study highlights the complementary strengths of PINNs and VQEs, as validated by implicit schemes. PINNs prioritize adher-

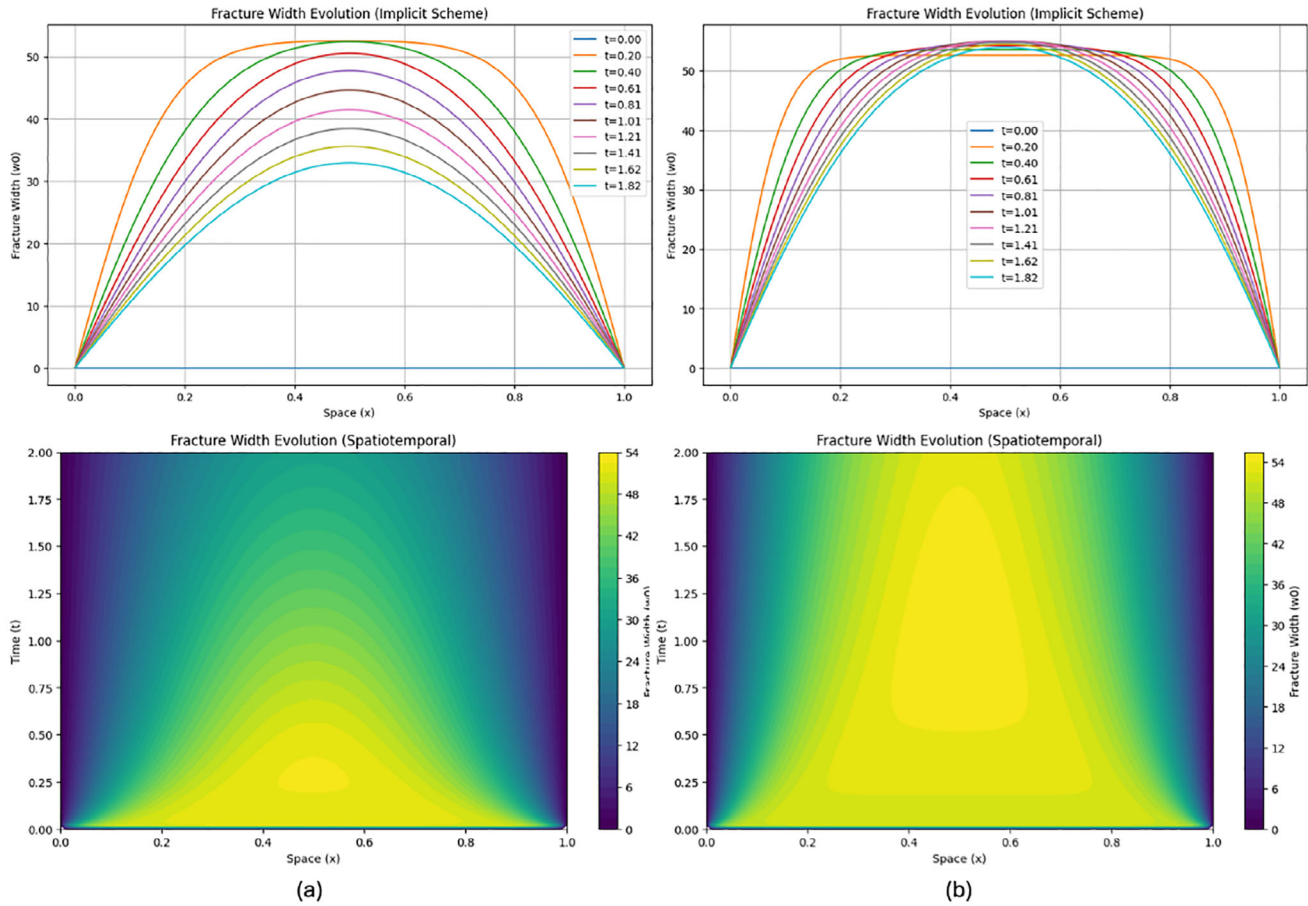


FIGURE 8 | Illustration of fracture-width evolution over space and time using the implicit scheme for two parameter sets. In column (a), $E' = 30.0$, $\mu = 5.0$, $H = 1.0$, $C_L = 1.0$, and $t_0 = 0.0$. The line plots show progressive widening with a smooth parabolic profile, and the contour plot shows a symmetric pattern with the maximum width near the domain center at later times. In column (b), $E' = 30.0$, $\mu = 8.0$, $H = 2.0$, $C_L = 1.0$, and $t_0 = 0.0$. The higher viscosity is associated with slower width evolution, as reflected by the more gradual profiles. Overall, the figure highlights the influence of viscosity and fracture height on simulated fracture evolution under the implicit reference model.

ence to physical constraints but can underestimate crack width for certain parameter regimes without further calibration. VQEs prioritize speed and computational efficiency while maintaining good qualitative agreement with the implicit baseline. Future research will explore hybrid architectures that combine the validation robustness of implicit schemes, the accuracy of PINNs, and the computational efficiency of VQEs. Such approaches can support the development of scalable decision-support tools for hydraulic-fracturing optimization in complex geological settings.

Author Contributions

Dennis Delali Kwesi Wayo: conceptualization, methodology, software, validation and visualization, writing – original draft, review and editing. **Sonny Irawan:** conceptualization, methodology, supervision, project administration, funding acquisition, writing – review and editing. **Mohd Zulkifli Bin Mohamad Noor:** methodology, validation, project administration, funding acquisition, writing – review and editing. **Mudasar Zafar:** methodology, validation and visualization, writing – review and editing. **Shynar Juziyeva:** methodology, software, validation and visualization, writing – review and editing. **Camila Martins Saporetto:** methodology, software, validation, writing – review and editing. **Leonardo Goliatt:** conceptualization, methodology, software, validation and visualization, writing – original draft, review and editing.

Randy Hazlett: conceptualization, supervision, project administration, funding acquisition, writing – review and editing. All authors reviewed the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data and code that support the findings of this study are openly available in the Nordgren-FracSim repository at <https://github.com/DennisWayo/Nordgren-FracSim> GitHub. The datasets used for training and the corresponding results are included in this article. Python scripts

for PINN (Google Colab) and VQE simulations (Qbraid) are available in the same repository.

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